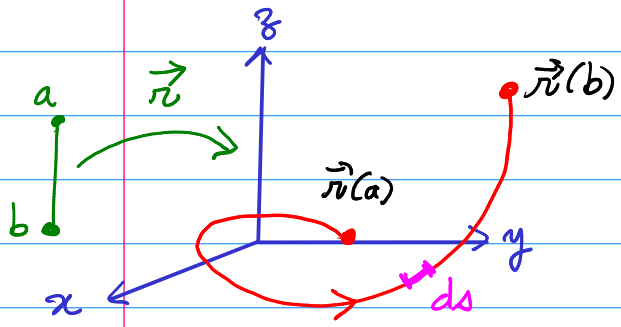
Surface Integrals

1) Surfaces (in 3D)

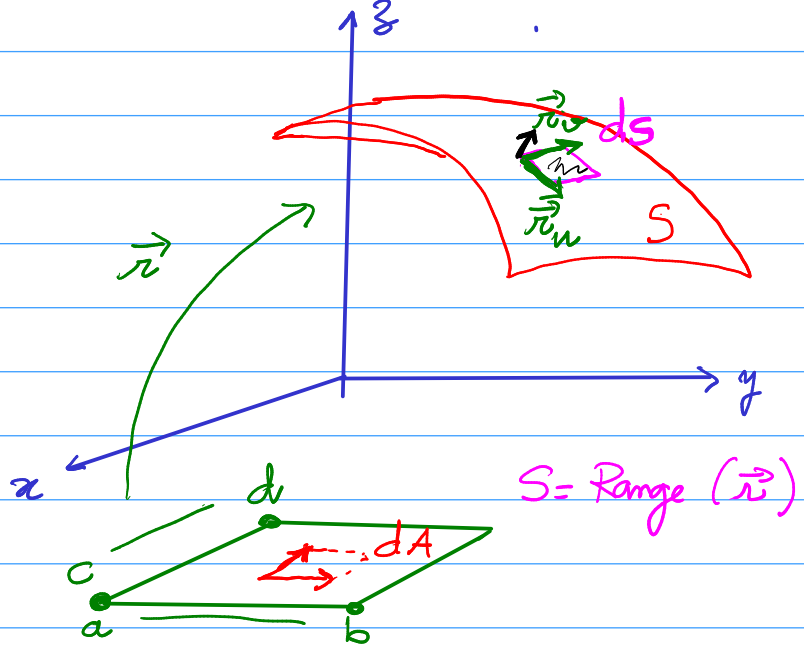
a) **Defn**: A surface in $\mathbb{R}^3$ is given by a parametrization

$$\vec{r} : [a;b] \times [c;d] \longrightarrow \mathbb{R}^3$$



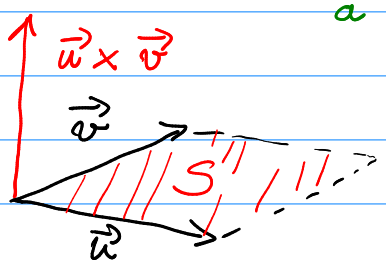
$$[a;b] \xrightarrow{\vec{r}} \mathbb{R}^3$$

$\mathcal{C}$ = Range $(\vec{r})$



S = Range $(\vec{r})$

b)



$$S = |\vec{u} \times \vec{v}|$$

Proposition: If $\vec{r} : [a;b] \times [c;d] \longrightarrow \mathbb{R}^3$

$(u, v) \longmapsto \vec{r}(u,v)$

then

$$\boxed{dS = |\vec{r}_u \times \vec{r}_v| \, dA}$$

Defn: The (surface) integral of $f : \mathbb{R}^3 \longrightarrow \mathbb{R}$ on the surface S = Range $(\vec{r})$ is given by

$$\iint_S f \cdot dS = \iint_{[a;b]\times[c,d]} f(\vec{r}(u,v)) \cdot |\vec{r}_u \times \vec{r}_v| \, dA$$

- The area of the surface S is given by

$$S = \iint_S dS = \iint |\vec{r}_u \times \vec{r}_v| \, dA .$$

2) Graph of a function as a surface

- S : graph of a function $f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}$

then $\vec{r} : D \longrightarrow \mathbb{R}^3$, $(x,y) \longmapsto \vec{r}(x,y)$ (Use $f(x,y)$)

$$\vec{r} = \langle x, y, f(x,y) \rangle$$
$$\vec{r}_x = \langle 1, 0, f_x \rangle$$
$$\vec{r}_y = \langle 0, 1, f_y \rangle$$

$$\Rightarrow \quad \vec{r}_x \times \vec{r}_y = \langle -f_x, -f_y, 1 \rangle$$

$$|\vec{r}_x \times \vec{r}_y| = \sqrt{1 + f_x^2 + f_y^2}$$

3) Examples : (a) $\iint_S x^2 \, dS$ on the sphere $x^2+y^2+z^2 = 1$

$\vec{r}(\varphi, \theta)$ $\begin{cases} x = \sin(\varphi)\cos(\theta) \\ y = \sin(\varphi)\sin(\theta) \\ z = \cos(\varphi) \end{cases}$

$$\vec{r}: [0;\pi]\times[0;2\pi] \longrightarrow \mathbb{R}^3$$
$$(\varphi,\theta) \longmapsto \vec{r}(\varphi,\theta)$$
$$|\vec{r}_\varphi \times \vec{r}_\theta| = |\sin(\varphi)| = \sin(\varphi)$$
(Check)

$$\iint_S x^2\,dS = \iint_{[0;\pi]\times[0;2\pi]} \sin^2(\varphi)\cos^2(\theta)\cdot\sin(\varphi)\,\underbrace{\sin(\varphi)\,d\varphi\,d\theta}_{dA}$$

?

Do you need a Jacobian here?

(b) $\iint_S y\,dS$ on $z = x + y^2$, $0 \le x \le 1$ and $0 \le y \le 2$

graph of a function

$$\Rightarrow z = f(x,y)$$
$$\Rightarrow |\vec{r}_x \times \vec{r}_y| = \sqrt{1 + 1^2 + 4y^2} = \sqrt{2+4y^2}$$
$$\iint_S y\,dS = \int_0^1\int_0^2 y\sqrt{2+4y^2}\,dx\,dy$$